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Interconnected learning between university and the workplace: a socio-cultural perspective of graduate employability in the UK

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ABSTRACT

Undergraduate students have been shown to spontaneously form communities of practice and social networks that support their learning and identity development. This study considers the translocation of this participatory learning into professional employment following graduation. Thirteen former biological sciences students from a single UK university were interviewed following their successful transition into graduate level scientific employment. A thematic analysis of the interview transcripts revealed that these students did not learn new workplace practices from more experienced peers *de novo*, rather they negotiated changes to their existing university practices and hence were able to participate as newcomers to the joint enterprise of their workplace community of practice. Hence, students' capacity for *know how* and *know who* knowledge has important implications for their employability, and this should be encouraged by maximising students' exposure to co-participatory learning both within the university and in collaboration with local community projects.

KEYWORDS

Communities of practice; legitimate peripheral participation; employability; identity; workplace; higher education

Introduction

An employability skills agenda

James Callaghan (1976), then British Prime Minister, defined the goal of education, as 'to equip children to the best of their ability for a lively, constructive, place in society, and also to fit them to do a job of work'. In this phrase work was almost an afterthought, and, at that time, it seems that the government viewed education as largely about enriching society. However, just twenty years later the role of education had changed. There had been an expansion in higher education (HE) in the United Kingdom (UK) leading to the formation of a collective of life-long learners who the Dearing Report (1997) termed a *learning society* able to become the skilled workforce of the future. Hence, the economic role of universities had shifted from being providers of technology to producers of a qualified labour force (Grotkowska et al., 2015).

This new role for HE has led to a change in the relationship between universities and the UK state. Perceiving HE as a producer of graduates who have an economic value to the state has resulted in a governmental ‘tightening grip over universities’ (Tomlinson, 2012, p. 426). There is the expectation that HE develops curricula to teach students ‘the transferrable work readiness skills’ because there is a demand for ‘employees with high level skills’ and ‘organisations find it hard to recruit the skilled people they need’ (Department for Business, Innovation and Skills, 2015, pp. 10–11). Consequently, the current UK Teaching Excellence Framework (TEF) incorporates a quality metric for employability, and the Higher Education Funding Council for England (HEFCE) has provided targeted support for HE programmes to deliver ‘high level skills’ within science, technology, engineering and mathematics (STEM) courses (Department for Business, Innovation and Skills, 2009, p. 7). Furthermore, employability skills are now seen as underpinning professional development (Belt et al., 2010) and a graduate’s life-long learning is now understood as ‘active management to their employability’ (Tomlinson, 2012, p. 426) to meet the demands of industry and commerce within a global economy.

The response of higher education to the skills agenda

The HE employability literature exemplifies skills such as critical thinking as paramount (Artess et al., 2017), and provides evidence-based employability skills profiles (Saunders & Zuzel, 2010) together with well-developed employability models and excellent case studies that can be linked to curriculum design (Pegg et al., 2012; Advance HE, n.d.). However, despite the immense amount of time, resources and cost invested in curriculum development to encompass employability skills, there is a lack of evidence of taught employability skills being transferred into the workplace (Cranmer, 2006; Hinchcliffe & Jolly, 2011; Holmes, 2001, 2013), and, in this context it is also interesting to note that O’Donnell and Tobbell (2007, p. 327 quotation marks in original), when considering adult learners’ transition into higher education, showed that ‘although attempts were made by course tutors to “teach” certain study skills to students through dedicated classes, the students only truly began to make progress in the learning of these skills by actively engaging in them’. Going further, Cranmer (2006) has suggested that resources could be better spent elsewhere, and Holmes (2013, p. 543) has suggested that ‘institutional-level curriculum development interventions were rarely, if ever, directly based on empirical research’ with much of the research concerning employability skills and attributes based on surveying of various HE stakeholders to ascertain ‘“perceptions” of the respondents: none attempt[ed] to devise some form of objective measure’ (p. 546 italics and quotation marks in original).

While there has been a drive to embed employability skills into the curriculum, there is no agreed definition of employability. Clark et al. (2016) favoured the view of Yorke (2006, p. 8) who understood employability as ‘a set of achievements – skills, understandings and personal attributes – that make individuals more likely to gain employment and be successful in their chosen occupations, which benefits themselves, the workforce, the community and the economy’. This widely used statement encapsulates employability as possession of skills and attributes, but such *possession* is problematic in four ways: firstly, there are many skills and attributes that need to be possessed (Department for Business, Innovation and Skills, 2011; Holmes, 2013), but their relative importance is unstated;

secondly, some skills which graduates may see as important may not be included (Hinchcliffe & Jolly, 2011); thirdly, it is unclear how employers are intended to make sense of such extensive skills lists (Holmes, 2013); and fourthly, there is little consideration that employers may require certain skills which HE is unable to effectively embed into their curricula because they are difficult to teach and objectively measure out of context.

Graduate identity

With such deficits in the employability skills agenda, alternatives have been proposed. Holmes (2001) challenges not only the dominant skills agenda, but also its consequences, such as the lack of funding for alternative ways to consider employability, and policy discussions that are solely skills driven. Holmes' (2001, 2006, 2013) concern with skills relates to *performance* which is always context specific and involves complex interactions with others or with artefacts. Consequently, performance of a skill is difficult to measure objectively. Rather it is better considered as situated learning (Lave & Wenger, 1991) which occurs within a social historical context, and can be recognised as the emergence of new practices and identities. Specifically, what students do and how they do it is dependent on their current identity and linked to their previous, historical experiences, and when students are taught in the abstract to demonstrate an employability skill or skills, for example, via a skills portfolio, their development of a graduate identity is overlooked.

Hinchcliffe and Jolly (2011) developed the graduate identity model by seeking the voice of employers in a way that went beyond skills and attributes to investigate which graduate identities and experiences were of most interest. They identified four strands of graduate identity: (1) values such as personal ethics and cultural awareness; (2) intellect, not just in degree classification but by being able to show, for example, intellectual curiosity and reflection; (3) performance in being able to deliver and implement skills and intellect in the workplace; and (4) engagement through willingness to meet personal, employment and social challenges head on and to be 'outward looking' (p. 580). As Hinchcliffe and Jolly (2011) note, considering graduate identity in terms of employability does not mean having to 'translate each and every employer requirement into an employability requirement' (p. 580). Rather, employers spoke about preferring students who had placement experience together with life experiences outside the domain of the curriculum, and, in this respect, they were seeking graduates who had participated in different types of communities of practice (CoP). 'It is the experience, albeit limited, of a community of practice that enables an employer to assess those all-important interpersonal skills' (p. 581). In sum, this finding of an alternative to the current skills agenda has important implications in terms of HE resources, staff time and governmental funding. While placements do provide exposure to different CoP, they are often external to the university, can require much organisation and may be limited in availability (Little & Harvey, 2006).

An under-researched area is the contribution to employability of students' ongoing social learning which is known to operate within HE. Through utilising social learning theory, Orsmond et al. (2013) showed that undergraduate biological sciences students form CoP spontaneously within the university and that these operate alongside the formal curriculum as an important component of learning and identity development.

Furthermore, within these communities learning is 'based on situated negotiation and re-negotiation of meaning' (Wenger, 1998, p. 51) that involves mutual engagement, joint enterprise and a shared repertoire. Orsmond et al. (2013) also showed that biological sciences students developed social networks that Wenger et al. (2011, p. 9) have defined as a set of 'relationships, personal interactions and connections among participants who have a personal reason to connect'.

Learning within communities of practice

Lave and Wenger (1991) sought to critique the conventional view of learning whereby a learner *internalises* knowledge. They expressed the view that such learning leaves the 'nature of learner, of the world, and of their relations unexplored' (p. 47). Their alternative view of learning was as an aspect of 'changing participation in changing communities of practice' (Lave, 1996, p. 150) with such learning referred to as *situated* and having a defining characteristic called legitimate peripheral participation (LPP). It is not possible within the confines of this paper to consider the more subtle aspects of CoP, and their development, which have been previously discussed (Ryan, 2015). Rather, this study remains close to the seminal work of Lave and Wenger (1991), and Wenger (1998) due to their more detailed consideration of the trajectory that newcomers take within a community of practice.

LPP is grounded in Vygotsky's earlier socialisation framework as described by Handley et al. (2006). Subsequently, Lave and Wenger (1991, p. 49 quotation marks in original) took 'a "collectivist", or "societal"' interpretation of Vygotsky's zone of proximal development to recognise the social transformation of LPP whereby learning moves beyond a pedagogic process to include the 'conflictual nature of social practice'. While there is 'no place within a community of practice designated "the periphery"' (Lave & Wenger, 1991, p. 36 quotation marks in original), LPP can be seen as a way to understand 'the gradual process of fashioning relations of identity as a full practitioner' (p. 121) and to search within CoP for the interconnection that a person has with activities within their world. In this dynamic learning environment of changing identities, knowledge changes as a result of ongoing practice and is understood as *knowledgeability* (Lave, 2008).

CoP are also known to occur within workplaces (Wenger, 2009). Hence, graduating students moving into graduate employment must transit from their former university CoP to become newcomers in their workplace community (Fenton-McCreedy et al., 2015). Successful transition of students from their university into employment is not simply a matter of having relevant skills, but of 'ways of being and relating in professional contexts' (Goldie, 2012, p. e641); a view that resonates with that of Hager and Hodgkinson (2009) who recognised learning transfer as a metaphor, not a literal event. They proposed that learning should be thought of as 'becoming' (p. 633) within a transitional process of boundary crossing, with Wenger (2010a) noting that boundaries, such as those between universities and work, are important learning systems connecting two different CoP. In sum, it seems that fostering students' graduate identities to ensure their success in the workplace cannot be solely pedagogically driven, but must occur through LPP within a community of practice that involves 'becoming' and 'changing who we are' (Wenger, 1998, p. 5).

Purpose of this study

As outlined previously, the UK government's skills agenda is problematic and does not seem to produce the desired results. Hence, there is a gap between the preparation of undergraduate students for employment and the recognition of that preparation once they enter employment as graduates. This study, through a CoP approach within a social learning theory framework (Farnsworth et al., 2016), explores graduates' workplace practice, and seeks its origins within their previous undergraduate CoP. By doing this we hope to unmask the role of engagement in CoP and face-to-face learning networks within students' career preparation. We argue that recognising this participatory learning should lead to a shift from focussing HE resources on teaching skills towards the enrichment of students' informal social learning.

Hence, this study focuses on the relationship between the participatory learning (Sfard, 1998) that students adopted as undergraduates and that which they now use as graduates in work. Such learning can be recognised as identity development (Orsmond & Merry, 2017) which allows us to further develop the notion of graduate identity introduced by Holmes (2001, 2006) and refined by Hinchcliffe and Jolly (2011). By taking a CoP approach we argue that graduates may not directly transfer skills from university into their workplace, rather university learning must be translated through participation in their new workplace communities.

Hence, the specific aim of this study is to investigate whether undergraduate biology students develop employability practices within their university CoP that are translated into their workplace communities when they enter graduate-level employment. The following research question was addressed:

- What is the role of university participatory learning within CoP in preparing biology undergraduate students for the workplace?

Methodology

Setting

The study concerned graduates from the Biological Sciences Department of a single post-1992 UK University who were subsequently in graduate-level employment utilising their scientific expertise. They had experienced university curricula which had been accredited by academic external reviewers, the Health Care Professions Council, the Institute of Biomedical Science and the Royal Society of Biology, and hence their learning content was assumed to be typical of the sector.

At the time of interview the participants were based at 11 different sites of employment within the UK. The nature of their work is described below. Three participants (designated 7, 8, 9, see below) were based together, albeit in different laboratories, but the remainder worked in individual locations. The three participants sharing a common workplace were interviewed within that setting. All other participants were interviewed at their former university.

Participants

Thirteen participants were identified using a mixture of opportunistic and snowball sampling (Cresswell, 2008). They received no inducements to participate.

To maintain anonymity the participants were numbered 1–13. Five were employed as Biomedical Scientists (Participants 1, 7, 8, 9, and 13), four as science teachers (Participants 3, 4, 6, and 11), three as laboratory technicians (Participants 2, 5, and 12), and one as a commercial representative for healthcare products (Participant 10). The participants had graduated between one and eight years prior to the study (mean 3.7, S.D. 2.3) with Participants 1, 2, 7, 8 and 10 as the most recent graduates, and Participants 4, 5 and 9 as the most historic graduates.

Instruments

Individual semi-structured interviews (Cohen & Manion, 1994; Cresswell, 2008) were carried out in order to investigate the nature and the extent of participants' university learning translation into the workplace. The interview schedule primarily concerned:

- How their university experiences have contributed to their overall learning.
- How their workplace experiences contribute to their overall learning.
- What judgements they make concerning the worth of university and workplace learning.
- With whom they discuss their workplace learning.
- The roles of university and workplace in their development of a professional identity.

The interview schedule was developed as described by Fielding (1993). Briefly, this involved: firstly, identifying topics surrounding the research question; secondly, clustering and sequencing relevant topics; and, thirdly, designing informal interview probes.

Procedures

These were as previously described (Merry & Orsmond, 2020). Briefly, each participant took part in a single informal interview of approximately 45 minutes duration where the questioning was exploratory (Higgins et al., 2002) and encouraging to elicit underlying attitudes (Fielding, 1993). All interviews were confidential, and the data generated was anonymous.

Interviews were conducted in the presence of two interviewers (the first two authors) who made contemporaneous notes which they discussed and refined after each interview. The interviews were also audio recorded enabling verbatim transcription. While the authors were all tutors at the participants' former university, they and the transcriber had no involvement with the work practice of the participants.

Analysis

Qualitative analysis of participants' interview data involved previously described procedures (Merry & Orsmond, 2020). Briefly, all authors read the interview transcripts to produce a sense of the whole. Then units of meaning relevant to the research question

were identified which were subsequently clustered to identify general themes (Cohen & Manion, 1994). Finally, representative quotes were sought to illustrate each theme and, in the present study, these were then linked to Hinchcliffe and Jolly's (2011) graduate identity framework using the codes: **V** for values; **I** for intellect; **P** for performance; and **E** for engagement. Verification at each stage of the process was sought through discussions involving all authors and further interrogation of the interview data and contemporaneous notes as required.

The study documentation and procedures were approved by the Staffordshire University Research Ethics Committee.

Findings

This study addressed a single research question concerning the role of university participatory learning in preparing biology students for the workplace with the data being analysed through the lens of Wenger's (1998) model of CoP. From this perspective, the authors identified five themes emerging from the data. These themes concerned the three attributes of CoP described by Wenger (1998) i.e. mutual engagement, joint enterprise and shared repertoire, together with a broader theme concerning learning at boundaries of CoP, and a power and identity theme. These five themes are considered in turn and illustrated with representative quotations containing phrases that are linked to Hinchcliffe and Jolly's (2011) graduate identity framework using the codes **V**, **I**, **P** and **E**.

1. *Mutual engagement: negotiating meaning at university and at work*

When we were first years [at university] ... we started to learn [who] we could gravitate towards (I) ... as time goes on you start trusting people and their opinions (**V**) so you start ... extracting actually what is important to you (**P** and **E**) [from those] whose opinion you do value more ... it wasn't about friendship it was ... I suppose ... the equivalent of professional ... opinion (**V**). And that has held on during teaching [my current employment] as well, because there are different people to go to for different things (**P**). There are definitely people I've gone to over the last year for different things ... I suppose I'm just embracing their areas of strength. As a department that works well because we've got areas of strength ... you call on these strengths, you become strong (**E**). (Participant 6)

This quote shows that, at university, the participant encountered a variety of people and, recognising their different identities, became able to decide whom to go to, talk to or watch, and whom to trust. Knowledgeability is evident through relational negotiation in ongoing practice with 'diverse others participating differently' (Lave, 2008, p. 292). The quote also shows the recognition of individual competencies, and, in this process, the participant uses the strength of the CoP to become a stronger member and, through trust, to develop a sense of belonging.

More broadly, it was evident from all the participant interviews that talking to different people at university was important. The participants' social learning practices did not begin *de novo* when they began their employment, but had historical origins in their former university practices. In terms of LPP, they are not currently learning by absorbing the new workplace talk, but are learning how to talk in the workplace (Lave & Wenger, 1991). Through using practices developed at university it seems that

participants have become aware of the practices of others within their new workplace CoP. All the workplaces represented in this study appear to be viable CoP (Fuller et al., 2005) where mutual engagement supports community of practice coherence and success.

2. Joint enterprise: developing communal understanding

Talking to confident (**V**) students [at university] really helped a lot ... they were confident because, I realised (**I**), they were not scared to experiment ... [They said] we'll give it a try. I've taken that into the workplace [laboratory] ... Some ideas we've never done before [and] we don't know if they are gonna work ... [The workplace] asked for ideas how to improve the way we do things and I wrote down what I thought at the time we could help to improve (**V, I and E**) ... Five months later a manager came to me and said 'thank you very much, you've won yourself fifty pounds' ... for a very good idea ... So going for interviews I can say my idea was highly recommended, and at work they started to change things [to] the way we're doing them (**P**). (Participant 1).

This quote illustrates examples of joint enterprise (Wenger, 1998). Joint enterprise does not mean that everyone agrees or has a common fixed belief about what is important to the community of practice. Rather it relates to a communal understanding derived from mutual engagement where different points of view are recognised and have importance. While the current practice of the participant is seen to have a historical context derived from practices they learnt at university, there is also *imagination* (Wenger, 2010b) producing a sense of belonging to their new workplace community of practice and the confidence to *suggest ideas*. Socialisation within a community of practice can be understood as '*participation as a way of learning*' (Lave & Wenger, 1991, p. 95 italics in original) with LPP 'important for developing "constructively naïve" perspectives or questions' (p. 117 quotation marks in original) concerning ongoing activities. Such participation is particularly illustrated by this quote because the participant, through LPP, develops their graduate identity and challenges 'old knowledgeabilities existing in practice' (Lave, 2008, p. 293). The participant's new idea changed the community of practice and through such means CoP do not remain fixed. Practices are not just replicated, but adapted through individual agency which, in turn, influences the mutual engagement processes that determine what matters to the community.

3. Shared repertoire: resources for negotiating meaning

... when I first went there [workplace] I always listen[ed] to how someone with experience who's been there a number of years ... organises the day. I always observe as well ... how they do things (**V**) ... I'd develop a working structure (**I**) ... that then becomes a routine ... working closely with the teachers, [and] other support staff (**E**) ... Forward planning [was required] based on what [practical] bookings are coming ... [I would] read through the practical ... [and] devise a way of doing it (**P**) ... safely ... [by] developing strategies (**P**) and ways of working ... [Like with] the type of dissertation I did [at university where I] had to book lab time ... [and] order for that work.[I] had to plan (**I**) ... read through resources ... [and] speak to the technicians (**V**) ... I needed to be organised (**P and E**). (Participant 5)

Wenger (1998, p. 83) calls the shared resources of a community of practice a repertoire to 'emphasize both its rehearsed character and its availability for further engagement in practice'. Using the example of taught practical sessions, the quote shows how the shared repertoires of different CoP are involved in negotiation of meaning, and through *alignment* (Wenger, 2010b) of activities new identities will emerge. Furthermore, it seems that the participant's use of the shared repertoire of their current workplace community of practice has a historical context derived from their previous university learning. In recognising the workplace need to 'listen to how someone with experience ... organises the day' the participant seems to be adopting their historical 'speak to the technicians' university practice; a finding that resonates with the view of Little (2003 as cited in Cranmer, 2006) who argued that the initial mismatch of skills between employers and graduates is short term because, once in employment, graduates do develop the context-specific skills they need. Overall, it is apparent that learning involves not just participants, but shared repertoires, and the historical context means that these repertoires do not have the same meaning for all community members. Through coordinating different resources, new perspectives on practice and new meanings emerge.

4. Negotiating meaning within a learning network and at professional boundaries

[At university I talked to] my sister ... She had a bit of experience on the forensic side (**V**) ... I was doing forensic biology ... [and] it just helped to talk through with people ... I think that is one way I've learned quite well (**I**, **E** and **P**) ... [Now that I am working as a technician] I've got my husband ... He did a science degree (**V**) as well so I'll go and talk to him if there's something ... I need to know ... He usually knows ... or [will] research it for me (**E**). [I talk to teachers at work and] a lot of them have respect that you've got different skills (**V**) ... Our side is to set the experiments up and know the little things that work or don't work (**P**) and to know the equipment. They need us and talk about what can make their lessons better (**E**) ... and we need them ... if we don't know something [because] they can help us (**P**). (Participant 12).

The quote shows that situated learning occurs not only within the participant's current community of practice, but also through interactions at the boundary of that community with external learning networks or other CoP. The learning occurring here is based on information being translated across the community boundary both outwards and inwards and it reflects ways of learning utilised by biological sciences students (Orsmond et al., 2013). Such multidirectional information flow illustrates *brokering* to provide new possibilities for belonging (Wenger, 2010b) and negotiating meaning by engagement with different repertoires and enterprises (Wenger, 1998). The quote concerns the utilisation of equipment, which is a resource with a different historical context inside and outside the community, and the learning also seems to be enhanced by the mutual *respect* derived from the social capital (Putnam, 2000) of those involved in brokering. Hence the quote illustrates the development of *knowledgeability* through LPP occurring 'spatially, in motion, in relation to diverse others, participating differently' (Lave, 2008, p. 292).

5. Graduate identity and workplace power relationships

If people perceive me as a teacher now, they don't see me as ... a scientist (**V**). But I still see myself as a scientist who teaches science (**I**) ... If you've got a form at school, you're ... a form teacher (**V**). [Identity] changes within the school ... People see you as a teacher a lot of the time ... but I've still got my [university] lab coat ... and I still put that on ... I make the kids aware (**P**) we're all scientists and they say 'we're not scientists'. [I say] ... 'you are since you walked through that door!'**(E)**. (Participant 6).

While issues of power have been considered to be marginalised in situated learning theory (Coutu & Willmott, 2003), Lave and Wenger (1991, p. 36) understood legitimate peripherality as 'a complex notion implicated in social structures involving the relations of power'. More specifically, Alvesson and Willmott (2002) identified three ways by which power regulates identity within workplace CoP: managerial; cultural-communitarian; and quasi-autonomous. In the quote, the participant, who was shown in a previous quote to mutually engage within their community of practice, recognises that their school organisational role entails *managerial* challenges to their perceived identity as a scientist. However, there is a shared *cultural-communitarian* understanding of teaching that allows the participant to take individual *quasi-autonomous* measures which seem to find expression through "micro emancipation," ... when employees have greater scope for arranging their own schedules and working practices, albeit with the parameters ... set by others' (Alvesson & Willmott, 2002, p. 624 quotation marks in original). In this way the participant can re-establish their identity as a *scientist who teaches science*. This finding resonates with the view of Wenger (2010b, p. 190) who argues that within a community of practice the agency of participants allows negotiation of meaning 'which even the most effective power cannot fully subsume'. The participant's strong identity as a scientist was gained at university, and part of the shared repertoire of that wider scientific community has been retained in the form of a *laboratory coat* which is used as powerful imagery to self-identify when in dialogue with pupils. This illustrates the resilience of historical university identities derived within CoP, and their potential to shape professional workplace practice.

Overall, our study data provide strong evidence that students successfully transition into graduate employment through the translation of their university participatory practice into their workplaces where it serves to foster their learning and identity development in this new setting.

General discussion and conclusions

The aim of this study was to consider employability as the translation of students' learnt practices from university into their workplace CoP. We have used a robust implementation of social learning theory as suggested by O'Brien and Battista (2020), and have developed the work of Orsmond et al. (2013) that previously showed the contribution of CoP to the learning of undergraduate biological sciences students. From this study it is apparent that a key way by which universities influence the workplace is through providing graduates with existing social learning identities and practices including the ability to negotiate meaning. Students' transition into the workplace is a transformational process (Kyndt et al., 2017) providing an opportunity for them to negotiate new meanings and

develop new identities with the meanings ‘at once both historical and dynamic, contextual and unique’ (Wenger, 1998, p. 54), and, in this way, ‘our identities form trajectories, both within and across communities of practice’ (p. 154).

Specifically, our findings show four things:

- (1) Students develop employability practices and identities when participating within university CoP.
- (2) Students translate these identities and practices once they have transitioned into the workplace.
- (3) In the workplace former students negotiate changes to existing practices through LPP, and so their identity trajectories are modified.
- (4) The practices and identities that students develop at university and modify through workplace practice reflect the graduate attributes sought by employers in terms of values, intellect, performance and engagement (Hinchcliffe & Jolly, 2011).

Muhamad (2012, p. 883) has argued that *know what*, *know why*, *know how* and *know who* knowledge is required ‘to prepare people for working life’, but HE has conventionally focussed on developing students’ *what* and *why* knowledge, which is mainly codified, rather than *how* and *who* knowledge, which is mainly tacit and often developed outside the formal curriculum (Knipprath, 2017). Our data show that CoP have the potential to address this discrepancy because students’ observation of others and negotiation of meaning within and at the boundaries of CoP leads them to make judgements regarding best practice (i.e. *know how*) and of the expertise of others (i.e. *know who*).

More broadly, the current state of the graduate labour market has been summarised by Lauder and Mayhew (2020) who discuss graduates’ ‘struggle to find good jobs, while employers say they cannot find the people with the skills they need’ (p. 1). This situation has arisen despite ‘preparing students for their life beyond formal education appear[ing] to be one of the main tasks of education’ (Grosemans & Kyndt, 2017, p. 209), and ‘survey evidence suggest[ing] that in the UK more than one third of workers use only a proportion of their skills’ (Winterton & Turner, 2019, p. 543). Winterton and Turner (2019) go further by suggesting that ‘part of the difficulty in achieving [graduates’] work readiness stems from differences in perception between the worlds of education and work’ (p. 540). Despite the prevalence of the employability skills agenda in HE, James Callaghan’s (1976) view of the societal role of HE seems to remain strong in some quarters with views being expressed such as ‘universities should be about developing intelligent and ethical citizens rather than “oven-ready chickens”’ (Winterton & Turner, 2019, p. 537 quotation marks in original), and that ‘the UK regularly manifests chronic skills gaps and shortages in part because of employers’ reluctance to invest in training’ (p. 542). To investigate employer opinions, Hinchcliffe and Jolly (2011) surveyed 105 UK commercial and public sector organisations and found that most would be prepared to wait for a year after appointment before graduate employees were expected to a) suggest structural changes within the organisation to overcome weaknesses (i.e. *know how* knowledge), and b) undertake independent work and negotiate with others (i.e. *know who* knowledge). However, they did expect graduates to show potential in these areas when they were first employed.

It is worth noting that some of the more technical sectors, such as IT, engineering and medicine, have influential professional organisations which accredit graduates and university curricula, explicitly demonstrating their particular knowledge base (Hinchcliffe & Jolly, 2011). Since such curricula may also benefit from the targeted HEFCE funding for STEM subjects, *what* and *why* codified technical skills ‘related to the field of the workers’ (Md Nasir et al., 2011, p. 9) have tended to influence university-wide employability programmes. However, ‘[n]on-technical skills are necessary to balance the technical skills of a skilled worker’ (p. 15), and both types of skills can be considered *transferable* and ‘essentially job-related skills, but not job-specific’ (Chadha & Nicholls, 2006, p. 117). We argue that the graduate *skills gap* between the *know how* and *know who* skills that employers seek, and the *know what* and *know why* skills that graduates possess may underlie any *skills shortage* (Green & Ashton, 1992; Winterbotham et al., 2020) perceived by employers. These discrepancies are unlikely to be addressed within the current ‘instrumental view of education ... in the Anglosphere nations’ (Lauder & Mayhew, 2020, p. 1) where students’ relational learning within CoP is considered ‘invisible’ (Merry & Orsmond, 2020, p. 1320). Through recognising the value of CoP to develop students’ *know how* and *know who* knowledge, graduate identity and employability can be enhanced.

Implications for higher education

Our findings challenge the narrower employability focus of the UK government and HE on individualised *transferable skills* as they ignore the ongoing benefits of LPP within CoP. As students’ employability and career progression are related to their ability to modify their practice through LPP we strongly suggest that universities should encourage students’ recognition of this participatory learning as part of their professional development rather than perpetuating an employability skills agenda with, it seems, little impact. Furthermore, a refocus of HE resources and expenditure on participatory learning and identity development within CoP could enhance students’ learning within both HE and in their subsequent workplaces. We argue that HE has the potential to create a more effective graduate workforce by encouraging the development of the identities and practices that students are naturally gaining through their ongoing participatory learning. This might be done in three specific ways:

Firstly, so that the value of LPP is made explicit to both tutors and students, degree-level study should become considered as engaging in co-participatory learning processes and, to this end, tutors need to reconsider their assessment procedures. Both students (Orsmond et al., 2013) and tutors (Merry & Orsmond, 2020) do learn by LPP within academic CoP, but this *invisible* learning remains unrecognised (Merry & Orsmond, 2020; Orsmond & Merry, 2017). Previously, Wegner and Nückles (2015) have made the distinction between learning in terms of acquisition of a knowledge entity, which can be objectively measured, and learning through participation, which is recognised in what someone does and so requires a more subjective professional evaluation. Since both university and workplace learning involve enculturation into CoP, tutors should review their traditional assessment practices to include students’ change in learner identity (Merry & Orsmond, 2020). By making learning outcomes less narrowly prescribed, the room for judgment making increases. This enables students to recognise themselves as

co-participants in a subject discipline working alongside tutors providing tutorial opportunities as *assessment as learning* events (Earl, 2003) where feedback is provided, not on coursework, or examination, but on participation in practice. Furthermore, encapsulating this feedback as assessing value creation within CoP provides opportunities for *immediate, potential, applied, realised* and *reframing* effects on student community dialogue (Wenger et al., 2011). Importantly, this approach is not simply about mark allocations; it is about the recognition of being and becoming a professional through identity change and practice, and ensuring that the value universities place on students' learning accurately reflects its usefulness (Lave & Wenger, 1991).

Secondly, tutors should be encouraged to provide the type of feedback described above in response to a variety of activities and on multiple occasions. Emphasising the role of participatory learning as an important element of professional development could be achieved, for example, by considering the curriculum as dynamic in the way discussed by Fraser and Bosanquet (2006, p. 275) where co-participation 'emerges from the needs of the students and the interactions between students, teachers and colleagues'.

Finally, tutors and students should be encouraged to participate in the wider communities both within and surrounding their individual universities. By this means students could become involved with other CoP in the conduct of authentic assignments that have a direct influence on local communities. For example, biology and arts students could collaborate to undertake tasks to help develop derelict land areas as part of a civic community project, or to develop infographics or websites that explain vaccination to specific cultural communities.

Through these suggestions, students could develop graduate identity attributes desired by employers (Hinchcliffe & Jolly, 2011).

Limitations of the study and suggestions for further research

While this study has provided rich data by taking a novel approach to study an important topic, the participant number was small, they were all biological or biomedical graduates from a single university department, and they had all secured graduate-level employment within scientific disciplines where the expected codes of discourse and behaviour might be broadly similar to their university experiences. Furthermore, only continuing employees were interviewed, i.e. participants that had successfully engaged with their new workplace CoP. While the context of this study might be similar to other technical sectors, such as IT, engineering and medicine, where particular emphasis is placed on discipline-specific knowledge and capabilities gained at university (Hinchcliffe & Jolly, 2011), the broader implications of the data must be considered tentatively until subsequent studies have taken place to investigate to what extent these findings can be generalised. A larger cohort of participants, perhaps from multiple studies, would provide greater confidence in our findings and might allow worthy comparisons to be made between university curricula, university disciplines, types of employment and degrees of success in the workplace.

A further caveat in the interpretation of the study findings is that the observed effects of university learning on the workplace learning environment are the participants' perceptions of these effects. These data should be triangulated with observational findings to increase their worth.

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No potential conflict of interest was reported by the authors.

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